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Spacecraft Attitude Estimation Based on Star Tracker and Gyroscope Sensors

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***Abstract*—The Attitude Determination (Estimation) System is the most important component for any spacecraft. It is the process of estimating the orientation of the spacecraft which cannot be measured directly from the sensors. This paper develops two attitude estimation techniques that rely on star tracker measurements rather than on rate gyro measurements. However, gyros have an error due to drifting (bias), meaning that their measurement error increases with time. An Extended Kalman Filter (EKF) is presented to compensate this bias and estimate the attitude of satellite. The results show the performance of the presented algorithms through simulations.**

Keywords—extended Kalman filter, star tracker, gyros attitude estimation, gyroscopes' bias, satellite.

I. INTRODUCTION

The gyro stellar is a combination of a star tracker and gyros that provides the attitude and the angular rate information. This sensor is used today on modern spacecraft to determine accurately the satellite attitude [1] and [2]. The measurement of these sensors can be easily integrated in order to estimate the orientation parameters (attitude and rotation rates) of the satellite precisely [3] and [4]. However, gyros have an error due to drifting (bias) [17] and [18], meaning that their measurement error increases with time. This bias term deteriorate the filter efficiency, the attitude data accuracy and causes the filter diverge in long terms.

The common approach is to compensate the integration drift of gyros by importing the external reference information obtained from attitude determination sensors such as star tracker. The bias caused by gyros could be compensated by introducing the output of these sensors in the estimator filter, because there is no drift accumulation in these sensors [6] and [8]. Compared with the other devices, the star tracker is advantageous for its high precision, solid state, high stable and wide workable range.

To compensate the negative effect on the performance of gyroscope caused by the bias, many approaches have been proposed [10],[17] and [18]. In this paper, the attitude and gyro bias estimation algorithm is presented as a part of the attitude determination procedure for a small satellite to estimate parameters, which cannot be measured directly from the sensors.

To meet the requirements of high pointing accuracy of the spacecraft, there are many attitude estimation algorithms based on state estimation. The extended Kalman filter (EKF) method is widely used in nonlinear systems [9],[15] and [16]. EKF which has not so many implantation examples for the gyroscopes' bias estimation in literature is used to determine the accurate attitude and rotation rate in three axes of satellite. In order to enable estimation of all parameters simultaneously without a high computational cost, two estimation cases are presented and discussed:

Case 1: Quaternion and Angular rate Estimation State using star tracker only;

Case 2: Quaternion and Gyros' bias Estimation State using gyro stellar.

This paper is composed of five sections as follows. Section 2 presents the dynamic and kinematic models used for the satellite. In the next section, we describe the model of star tracker and gyros sensors. Through section 4, the description of the Kalman estimator is presented. Finally, the paper is concluded by testing the presented cases and discusses their significance via simulations.

II. PRELIMINARIES

In this section, a brief review of the dynamic and kinematic equations of satellite motion is shown.

A. Attitude Dynamics

The dynamic of a spacecraft in inertial space is governed by Euler's equation of motion [11], [12] and [13]

$$\mathbf{I}\dot{\boldsymbol{\omega}}_{\mathrm{B}}^{\mathrm{I}} = \mathbf{N}_{\mathrm{GG}} + \mathbf{N}_{\mathrm{D}} + \mathbf{N}_{\mathrm{MT}} - \boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{I}} \times \left(\mathbf{I}\boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{I}} + \mathbf{h}\right) - \dot{\mathbf{h}}] \qquad (1)$$

Where: $\boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{I}}$, $\mathbf{I}$, $\mathbf{N}_{\mathrm{GG}}$, $\mathbf{N}_{\mathrm{MT}}$, $\mathbf{N}_{\mathrm{D}}$ are respectively the inertially referenced body angular velocity vector, Inertia matrix of the spacecraft, gravity gradient torque vector, applied magnetorquer control firing, and Disturbance torque vector.

B. Attitude Kinematics

The spacecraft kinematic equations of motion in terms of quaternions can be updated by the following vector equation:

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega q} \tag{2}$$

$$\mathbf{\Omega} = \begin{bmatrix} 0 & \omega_{oz} & -\omega_{oy} & \omega_{ox} \\ -\omega_{oz} & 0 & \omega_{ox} & \omega_{oy} \\ \omega_{oy} & -\omega_{ox} & 0 & \omega_{oz} \\ -\omega_{ox} & \omega_{oy} & -\omega_{oz} & 0 \end{bmatrix} \tag{3}$$

$$\mathbf{q} = [q_1 \;\; q_2 \;\; q_3 \;\; q_4]^T \tag{4}$$

$$\boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{O}} = [\; \omega_{Bx}^{O} \;\; \omega_{By}^{O} \;\; \omega_{Bz}^{O} \;]^T \tag{5}$$

where: $\mathbf{q}$ and $\boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{O}}$ are respectively the attitude quaternion of satellite and the body angular rate vector in orbit coordinates.

The attitude transformation matrix of every vector of the orbital reference to coordinate body in terms of quaternions is expressed as follows:

$$\mathbf{A} = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1q_2 + q_3q_4) & 2(q_1q_3 - q_2q_4) \\ 2(q_1q_2 - q_3q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_2q_3 - q_1q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \tag{6}$$

From (3) and (5), the dynamic and kinematic models used for the satellite is expressed as:

$$\dot{q}_1 = \omega_{oz}\, q_2 - \omega_{oy}\, q_3 + \omega_{ox}\, q_4 \tag{7.1}$$

$$\dot{q}_2 = -\omega_{oz}\, q_1 + \omega_{ox}\, q_3 + \omega_{oy}\, q_4 \tag{7.2}$$

$$\dot{q}_3 = \omega_{oy}\, q_1 - \omega_{ox}\, q_2 + \omega_{oz}\, q_4 \tag{7.3}$$

$$\dot{q}_4 = -\omega_{ox}\, q_1 - \omega_{oy}\, q_3 - \omega_{oz}\, q_4 \tag{7.4}$$

$$\dot{\omega}_x = \frac{N_x}{I_x} - \alpha\omega_y\omega_z - \frac{h_z}{I_x}\omega_y + \frac{h_y}{I_x}\omega_z - \frac{\dot{h}_x}{I_x} \tag{7.5}$$

$$\dot{\omega}_y = \frac{N_y}{I_y} - \beta\omega_x\omega_z - \frac{h_x}{I_y}\omega_z + \frac{h_z}{I_y}\omega_x - \frac{\dot{h}_y}{I_y} \tag{7.6}$$

$$\dot{\omega}_z = \frac{N_z}{I_z} - \gamma\omega_x\omega_y - \frac{h_y}{I_z}\omega_x + \frac{h_x}{I_z}\omega_y - \frac{\dot{h}_z}{I_z} \tag{7.7}$$

Where, $\alpha = \frac{I_z - I_y}{I_x}$, $\beta = \frac{I_x - I_z}{I_y}$, $\gamma = \frac{I_y - I_x}{I_z}$

$$\mathbf{N} = \mathbf{N}_{\mathrm{GG}} + \mathbf{N}_{\mathrm{D}} + \mathbf{N}_{\mathrm{MT}} = [\mathrm{N_x}, \mathrm{N_y}, \mathrm{N_z}]^T$$

III. Measurement Sensor Models

A. The Star Tracker Model

This sensor provides the quaternion directly. The mathematical model of the sensor is [20]:

$$\mathbf{q}_{\mathrm{meas}} = \mathbf{q} + \boldsymbol{\eta}_1 \tag{8}$$

Where, $\mathbf{q}_{\mathrm{meas}}$ is the measured quaternion of the star tracker, $\mathbf{q}$ is the true quaternion satellite, and $\boldsymbol{\eta}_1$ is the zero mean Gaussian white noise with covariance matrix $\mathbf{R}_{\mathrm{s}}$.

$$\boldsymbol{\eta}_1 = N\{0, \mathbf{R}_{\mathbf{s}}\}$$

B. The Gyro Model

The advantage of a gyro is that it can provide the angular rates of the roll, pitch, and yaw of the satellite directly. The mathematical model of the sensor is [10],[14] and [17]:

$$\boldsymbol{\omega}_{\mathrm{BI,meas}} = \boldsymbol{\omega}_{\mathrm{BI}} + \mathbf{b}_{\mathbf{g}} + \boldsymbol{\eta}_2 \tag{9}$$

Where: $\boldsymbol{\omega}_{\mathrm{BI,meas}}$ is the measured angular rate of the gyroscope, $\boldsymbol{\omega}_{\mathrm{BI}}$ is the true angular rate. $\mathbf{b}_{\mathrm{g}}$ is the gyro bias vector, and $\boldsymbol{\eta}_2$ is the zero mean Gaussian white noise with covariance matrix $\mathbf{R}_{\mathrm{g}}$.

$$\boldsymbol{\eta}_2 = N\{0, \mathbf{R}_{\mathbf{g}}\}$$

IV. EKF Estimation

To describe all the details of the EKF is beyond the scope of this paper. Therefore, we present a more algorithmic description omitting some theoretical considerations.

In this work, EKF is used as the estimation algorithm in the nonlinear observer system. The two estimation techniques that are discuted in this section:

A. CASE 1: Inertial Angular Rate Quaternion EKF:

In the first case, the state vector (7 elements) to be estimated includes the inertial angular rates $\boldsymbol{\omega}_{\mathrm{BI}}$ and the attitude quaternion vector $\mathbf{q}$ using only the star tracker measurement. The full state vector can be represented as,

$$\boldsymbol{X} = [\boldsymbol{\omega}_B^I \; \boldsymbol{q}]^T = [\omega_x \; \omega_y \; \omega_z \;\; q_1 \; q_1 \; q_3 \;\; q_4]^T \tag{10}$$

1) A.1. Propagate state:

The state propagation by numerical integration can be seen in Equation (11) and (12) respectively:

$$\bar{\mathbf{q}}_{\mathrm{k+1}} = \hat{\mathbf{q}}_{\mathrm{k}} + \int_{t_k}^{t_{k+1}} (\mathbf{\Omega q})\mathrm{dt} \tag{11}$$

$$\bar{\boldsymbol{\omega}}_{\mathrm{k+1}} = \hat{\boldsymbol{\omega}}_{\mathrm{k}} \int_{t_k}^{t_{k+1}} \mathrm{I}^{-1}[\mathbf{N}_{\mathrm{GG}} + \mathbf{N}_{\mathrm{D}} - \boldsymbol{\omega}\mathbf{I}\boldsymbol{\omega}]\,\mathrm{dt} \tag{12}$$

2) Propagation Cycle:

- Covariance Propagation is :

$$\bar{\mathbf{P}}_{\mathrm{k/k+1}} = \boldsymbol{\phi}_{\mathrm{k/k+1}} \hat{\mathbf{P}}_{\mathrm{k/k}} \boldsymbol{\phi}_{\mathrm{k/k+1}}^T + \mathbf{Q}_{\mathrm{k+1}} \tag{13}$$

$$\bar{\boldsymbol{\phi}}_{k/k+1} = \mathbf{I}_{7\mathrm{X}7} + \mathbf{F}_{\mathrm{k/k+1}}\, \mathrm{T_s} \tag{14}$$

$$\bar{\boldsymbol{\phi}}_{\frac{k}{k}+1} = \mathbf{I}_{7\mathrm{X}7} + \begin{bmatrix} \left.\frac{\partial\dot{\omega}}{\partial\omega}\right|_{t=t_k} & \left.\frac{\partial\dot{\omega}}{\partial q}\right|_{t=t_k} \\ \left.\frac{\partial\dot{q}}{\partial\omega}\right|_{t=t_k} & \left.\frac{\partial\dot{q}}{\partial q}\right|_{t=t_k} \end{bmatrix} \mathrm{T_s} \tag{15}$$

The process noise covariance matrix Q by (ref above)

$$\mathbf{Q} \approx \begin{bmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{bmatrix} \tag{16}$$

Where,

$$\boldsymbol{Q}_{11} = diag[\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3, \mathbf{Q}_4]^T$$
$$\boldsymbol{Q}_{12} = diag[\mathbf{0}]_{4x4}$$
$$\boldsymbol{Q}_{21} = diag[\mathbf{0}]_{3x3}$$
$$\boldsymbol{Q}_{22} = diag[\mathbf{Q}_5, \mathbf{Q}_6, \mathbf{Q}_7]^T$$

$$\mathbf{Q}_1 = \frac{\sigma^2}{12}\left(\frac{q_4^2}{I_x^2} + \frac{q_3^2}{I_y^2} + \frac{q_2^2}{I_z^2}\right)\mathrm{T}_s^3,\ \mathbf{Q}_2 = \frac{\sigma^2}{12}\left(\frac{q_3^2}{I_x^2} + \frac{q_4^2}{I_y^2} + \frac{q_1^2}{I_z^2}\right)\mathrm{T}_s^3$$
$$\mathbf{Q}_3 = \frac{\sigma^2}{12}\left(\frac{q_2^2}{I_x^2} + \frac{q_1^2}{I_y^2} + \frac{q_4^2}{I_z^2}\right)\mathrm{T}_s^3,\ \mathbf{Q}_4 = \frac{\sigma^2}{12}\left(\frac{q_1^2}{I_x^2} + \frac{q_2^2}{I_y^2} + \frac{q_3^2}{I_z^2}\right)\mathrm{T}_s^3$$

$$\mathbf{Q}_5 = \frac{\sigma^2}{I_x^2}T_s, \mathbf{Q}_6 = \frac{\sigma^2}{I_y^2}T_s, \mathbf{Q}_7 = \frac{\sigma^2}{I_z^2}T_s$$

The partial derivatives of the system equation Eq.(11-12) against the state vector Eq. (10) are introduced and detailed in[19] and [20].

3) Correction Cycle

- Observation Matrix Computation :

$$\boldsymbol{Hs} = \left[\frac{\partial \mathbf{q}_{meas}}{\partial \boldsymbol{\omega}_{\mathrm{BI}}}\ \frac{\partial \mathbf{q}_{meas}}{\partial \mathbf{q}}\right] = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (17)$$

- Kalman Gain Matrix

$$\mathbf{K}_{k+1} = \overline{\mathbf{P}}_{k+1/k}\mathbf{Hs}_{k+1}^T(\mathbf{Hs}_{k+1}\overline{\mathbf{P}}_{k+1/k}\mathbf{Hs}_{k+1}^T + \mathbf{Rs}_{k+1})^{-1} \quad (18)$$

Update covariance:

$$\widehat{\mathbf{P}}_{k/k+1} = [\mathbf{I}_{7x7} - \mathbf{K}_{k+1}\mathbf{Hs}_{k+1}]\overline{\mathbf{P}}_{k+1/k} \quad (19)$$

- Update State

$$\widehat{\mathbf{X}}_{K+1/K+1} = \overline{\mathbf{X}}_{K+1/K} + \mathbf{K}_{k+1}\left(\mathbf{q}_{\mathrm{meas}} - \mathbf{Hs}\,\overline{\mathbf{X}}_{K+1/K}\right) \quad (20)$$

B. Quaternion and Bias EKF:

In the second case, the EKF algorithm for the gyros bias and attitude estimator is similar to the algorithm presented in case 1 except for a few differences, which will be mentioned below. The state vector (7 elements) to be estimated includes the attitude quaternion vector $\boldsymbol{q}$ and gyroscope's bias vector $\mathbf{b}_{\mathrm{g}}$, using both the star tracker and gyro measurement. The full state vector can be represented as:

$$\boldsymbol{X} = \left[\boldsymbol{q}\ \mathbf{b}_{\mathrm{g}}\right]^T = \left[\ q_1\ q_2\ q_3\ \ q_4\ b_{gx}\ b_{gy}\ \ b_{gz}\ \right]^T \quad (21)$$

1) Propagate state:

Only the non-linear kinematic and bias equations are propagate because in this case the angular rates are directly obtained from the gyros.

$$\overline{\mathbf{q}}_{k+1} = \widehat{\mathbf{q}}_k + \int_{t_k}^{t_{k+1}}(\boldsymbol{\Omega}\mathbf{q})\mathrm{d}t \quad (22)$$

$$\overline{\mathbf{b}}_{gk+1} = \hat{\mathbf{b}}_{\mathbf{gk}} + \boldsymbol{\eta}_2 \quad (23)$$

2) Propagation Cycle:

- Covariance Propagation is:

$$\overline{\boldsymbol{\Phi}}_{k/k+1} = \mathbf{I}_{7X7} + \begin{bmatrix} \left.\frac{\partial \dot{\mathrm{q}}}{\partial \mathrm{q}}\right|_{t=t_k} & \left.\frac{\partial \dot{\mathrm{q}}}{\partial \mathrm{b}_g}\right|_{t=t_k} \\ \left.\frac{\partial \dot{\mathrm{b}}_g}{\partial \mathrm{q}}\right|_{t=t_k} & \left.\frac{\partial \dot{\mathrm{b}}_g}{\partial \mathrm{b}_g}\right|_{t=t_k} \end{bmatrix} \mathrm{T}_s \quad (24)$$

3) Correction Cycle

- Observation Matrix Computation :

$$\mathbf{Hsg} = \left[\frac{\partial \mathbf{q}_{\mathrm{meas}}}{\partial \mathbf{q}}\ \frac{\partial \mathbf{q}_{\mathrm{meas}}}{\partial \mathbf{b}_{\mathbf{g}}}\right] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (25)$$

- Kalman Gain Matrix

$$\mathbf{K}_{k+1} = \overline{\mathbf{P}}_{k+1/k}\mathbf{Hsg}_{k+1}^T\left(\mathbf{Hsg}_{k+1}\overline{\mathbf{P}}_{k+1/k}\mathbf{Hs}_{k+1}^T + \mathbf{Rs}_{k+1}\right)^{-1} \quad (26)$$

- Update covariance

$$\widehat{\mathbf{P}}_{k/k+1} = [\mathbf{I}_{7x7} - \mathbf{K}_{k+1}\mathbf{Hsg}_{k+1}]\overline{\mathbf{P}}_{k+1/k} \quad (27)$$

- Update State

$$\widehat{\mathbf{X}}_{K+1/K+1} = \overline{\mathbf{X}}_{K+1/K} + \mathbf{K}_{k+1}\left(\mathbf{q}_{\mathrm{meas}} - \mathbf{Hsg}\,\overline{\mathbf{X}}_{K+1/K}\right) \quad (28)$$

- Update angular rate

Since the angular rate measurements are directly obtained from the gyro, the need to propagate angular rates is eliminated .The angular rates are obtained using the gyro measurements subtracted from gyro's bias estimated.

$$\widehat{\boldsymbol{\omega}}_{k/k+1} = \boldsymbol{\omega}_{\mathrm{BI,meas}_{k+1}} - \hat{\mathbf{b}}_{\mathbf{g}\,k/k+1} \quad (29)$$

V. Simulation Results:

The results presented in this section were obtained with a simulator that implements the dynamics of the satellite, to propagate expected angles and rates using C code, MATLAB, and SIMULINK. The simulator incorporates the star tracker and gyros as sensors to estimate: the attitude, gyros bias and angular rate of the satellite. The attitude estimators were run for two orbits. A summary of operating constants is presented in table 1.

TABLE I. Satellite Simulation Parameters

Parameters [units]	Value
Dimensions shape (m^3)	0.65 x 0.65 x 0.6
Satellite inertia matrix ($kg.m^2$)	$\begin{bmatrix} 40.45 & 0 & 0 \\ 0 & 42.09 & 0 \\ 0 & 0 & 40.36 \end{bmatrix}$
Inclination of orbit (deg)	98.2
Altitude (km)	680
Weight (Kg)	90
Initial attitude rate (deg/sec)	[0 -0.06 0]
Measurement Noise Covariance Matrix of star tracker **Rs**	10^{-9}

A. *Case 1:*

Fig. 1 presents the attitude error in the roll, pitch, and yaw axis of the satellite.

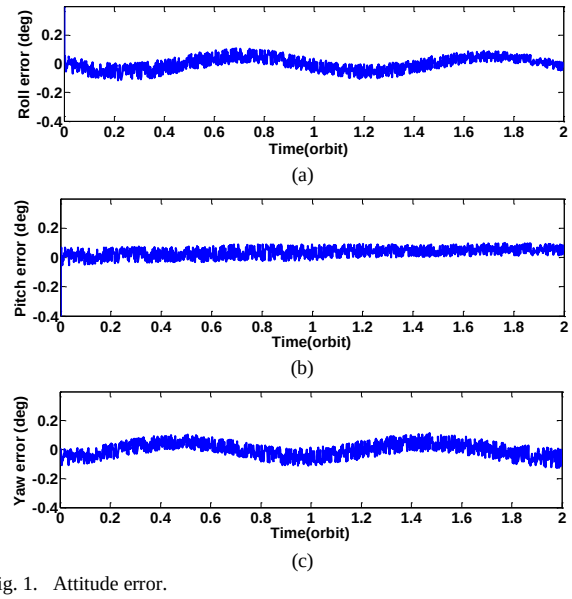

Fig. 1. Attitude error.

Fig. 2 shows the angular rates errors .

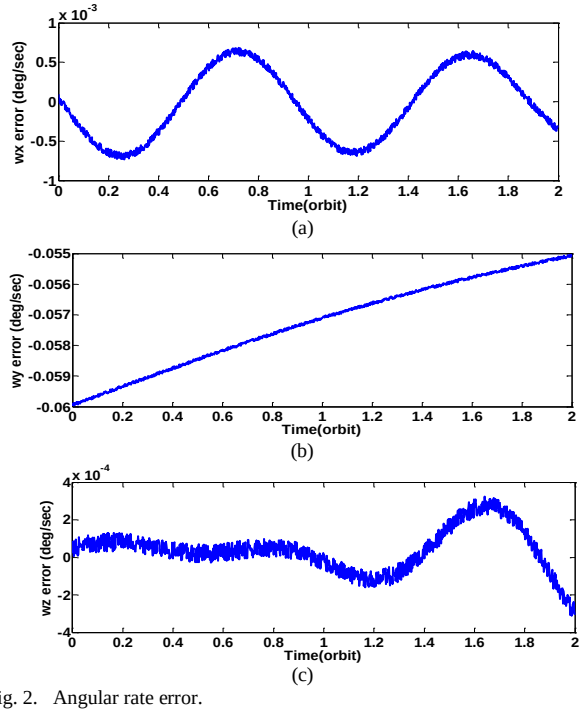

Fig. 2. Angular rate error.

Figs. 3 and 4 show the state vector covariance and the observability of the system respectively.

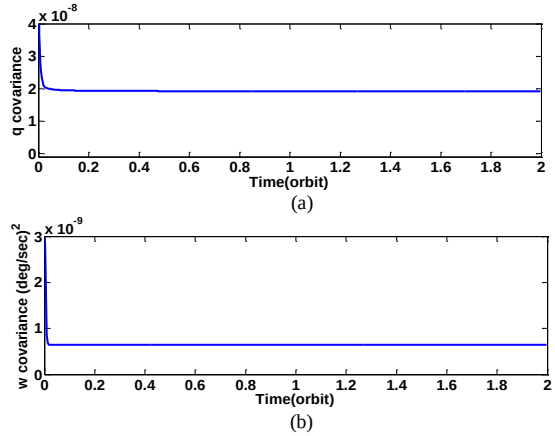

Fig. 3. The state covariance vector.

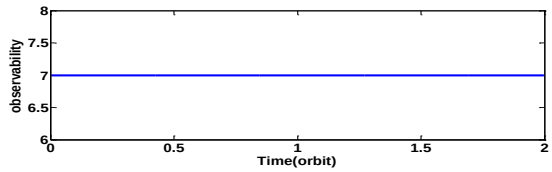

Fig. 4. The observability of the system.

B. *Case 2:*

Since the angular rate measurement is directly obtained from the gyro, the attitude estimate is expected to be more accurate. The attitude errors are shown in Fig. 5.

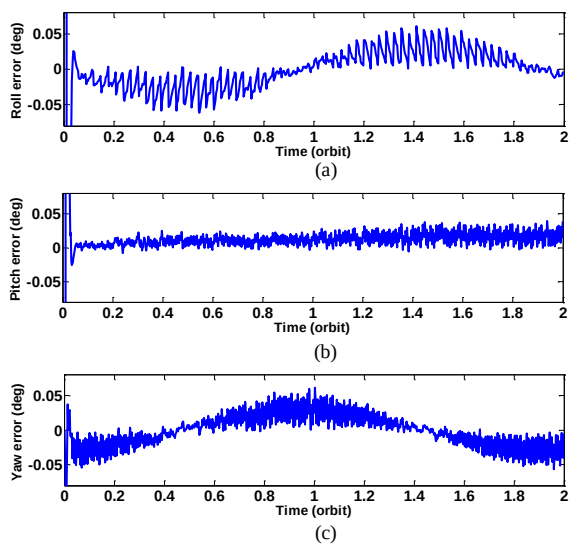

Fig. 5. Attitude error.

The actual and estimated gyros bias components as well as bias error components are shown in Fig.6 and Fig.7 respectively. Once the bias components are estimated they are subtracted from the gyros measurements which eliminates the actual bias effect.

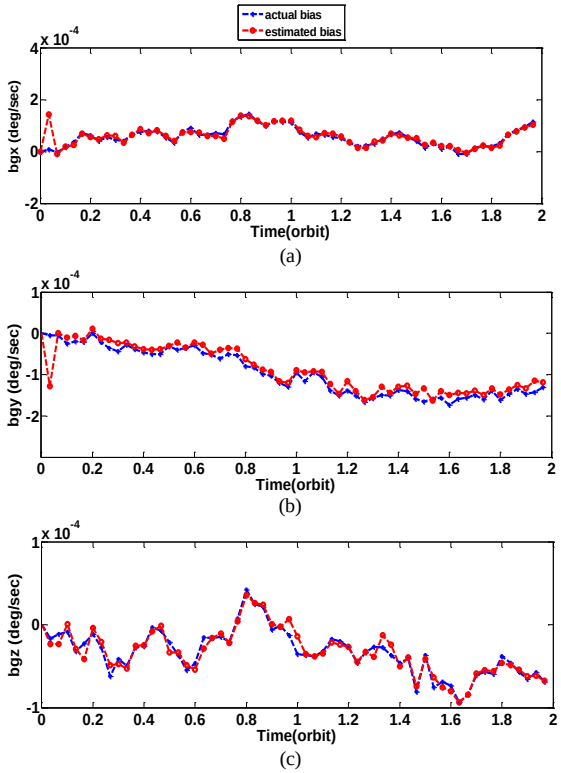

Fig. 6. Actual and estimated gyros bias.

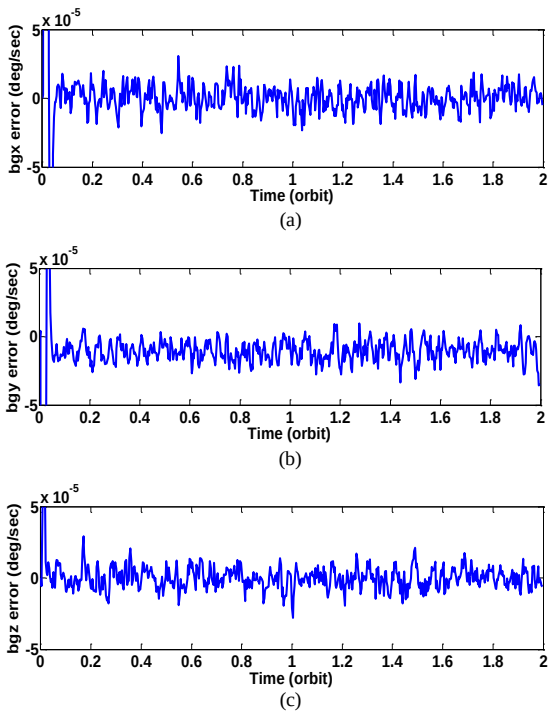

Fig. 7. Bias error.

Fig. 8 shows the plot error of the estimated angular velocities when compared to the real angular velocities measured by gyroscope respectively around X-axis, Y-axis, and Z-axis.

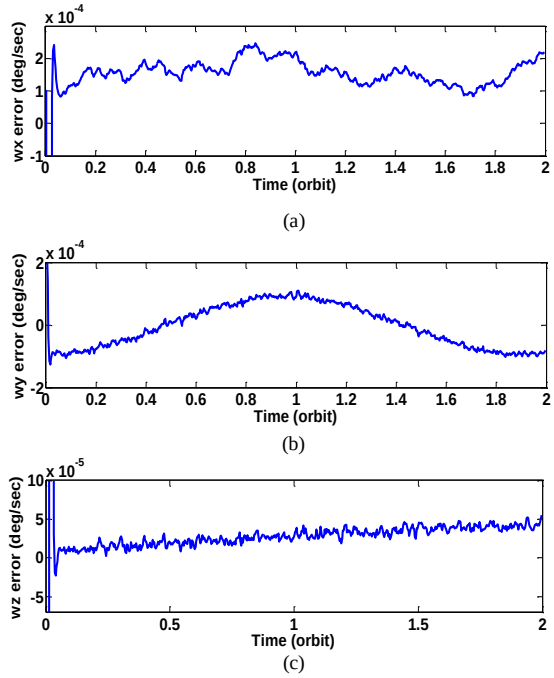

Fig. 8. Angular rate error.

Figs. 9 and 10 show the state vector covariance and the observability of the system respectively.

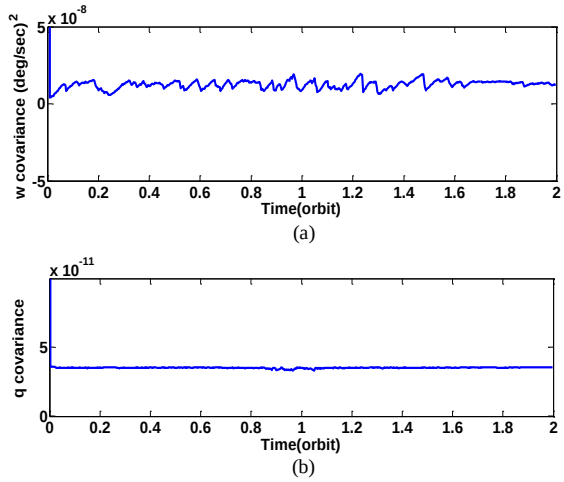

Fig. 9. The state covariance vector.

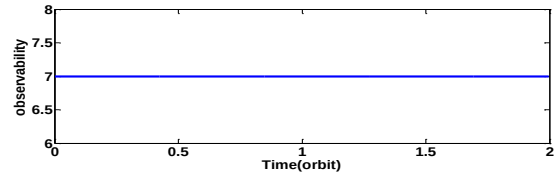

Fig. 10. The observability of the system.

TABLE II. ATTITUDE ERROR ESTIMATION COMPARISON

	RMSE case 1	RMSE case 2
Roll(deg)	0.0463	0.0242
Pitch(deg)	0.0472	0.0187
Yaw(deg)	0.0488	0.0235
Magnitude of error	0.0822	0.0386
Wx (deg/sec)	$2.2*10^{-3}$	$0.54*10^{-4}$
Wy (deg/sec)	$5.94 *10^{-2}$	$1.35*10^{-4}$
Wz (deg/sec)	$4 *10^{-3}$	$0.54*10^{-4}$
Magnitude of error	$5.89*10^{-2}$	$1.89*10^{-4}$

The Root Mean Squared Error (RMSE) in the two cases is shown in Table 2. The result shows that it is possible to determine accurately the attitude and angular rates much better when using case.2 estimator. The average RMS error in attitude was calculated to be less than 0.04 (deg) while the error in the angular rate estimates is $2*10^{-4}$ (deg/sec), because the angular rate measurements are directly available in this case whereas angular rates were estimated in the case.1

VI. CONCLUSION

In this paper, attitude determination was performed to estimate the quaternion and angular rates from star tracker and gyros data. Two separates EKF techniques were performed and discussed. The first case estimates quaternion attitude and the angular rates. The second case makes use of gyros measurements thereby eliminating the need to estimate the angular rates, however this case estimates the bias vector on the gyros along with the quaternion attitude. The results clearly indicate that when estimating the bias, attitude quaternion and angular rate estimates are more accurate.

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